

Module 15: Ecology

1. Define ecology. How does it differ from environmental science?
2. List the levels of ecological organization from smallest to largest (organism → biosphere). Give a brief description of each level.
3. What is a population? What factors affect population size (birth rate, death rate, immigration, emigration)?
4. Describe the difference between exponential growth and logistic growth. What is carrying capacity (K)?
5. What is a community? How do species within a community interact?
6. Define and give an example of each type of species interaction:
 7. Competition
 8. Predation
 9. Mutualism
 10. Commensalism
 11. Parasitism
12. What is a food chain? How does it differ from a food web?
13. Explain the 10% rule of energy transfer between trophic levels. Why are there rarely more than four or five trophic levels?
14. What is a keystone species? Give an example and explain why its removal would dramatically change the community.
15. Define ecosystem. What is the difference between biotic and abiotic factors? Give three examples of each.
16. Describe the water cycle. What role do evaporation, transpiration, and precipitation play?

17. Describe the carbon cycle. How do photosynthesis and cellular respiration contribute to the cycling of carbon?
18. What is the greenhouse effect? How do human activities (burning fossil fuels, deforestation) amplify it?
19. What is biodiversity? Why is it important for ecosystem stability and human well-being?
20. List three major threats to biodiversity (habitat destruction, invasive species, climate change). Explain how each reduces biodiversity.